

# Flight Information Query System Report

Group number: 22

## 1. System Design

### 1.1 Overview

The Flight Information Query System is a data visualization tool developed using Processing, designed to help users efficiently query and analyze flight data. The system reads flight information from CSV files, provides multiple filtering options, and displays results through an interactive interface.

### 1.2 Core Features

#### 1.2.1 Data Loading & Storage

- Reads flight data from `flights100k.csv` and stores each record as a `Flight` class object
- `Flight` class contains key attributes: date, airline, origin/destination, departure/arrival times, flight status

#### 1.2.2 Data Filtering

Users can filter flights by:

- Origin/Destination: Fuzzy matching of city names (case-insensitive)
- Date Range: Supports flight date or arrival date filtering (MM/DD/YYYY format)
- Flight Status: On-time, delayed, or cancelled (mutually exclusive selection)

The `filterFlights()` method implements dynamic combination of multiple criteria.

#### 1.2.3 Interactive Interface

- Main Page (Page 1):
  - Top: Filter panel (input fields, status buttons, chart toggles)
  - Bottom: Scrollable flight list with hover tooltips
- Flight Details (Page 2):
  - Displays comprehensive flight information in sections
- Visualization Pages (4-5):
  - Heatmap (Page 4): Airport delay patterns
  - Route Map (Page 5): Flight paths (static images)

#### 1.2.4 User Interaction

- Input Fields: Keyboard support with focus indicators
- Buttons:
  - Status buttons (radio button behavior)
  - Chart toggles for visualization pages
- Scrolling: Mouse wheel controls flight list

## 1.3 Technical Implementation

- Date Handling: `addOneDay()` processes overnight flights
- Tooltips: Displays concise flight info on hover
- Error Handling: Provides defaults (e.g., "N/A" for empty distance fields)

## 2. Extended Features

### 2.1 Interaction Enhancements

- Dynamic Tooltips: Real-time flight summaries on hover
- Dual-Date Filtering: Supports both flight and arrival date ranges

### 2.2 Visualization Improvements

- Heatmap & Route Map: Presents analyzed data visually
- Multi-page Navigation: Clear functional segmentation

### 2.3 Real-time Statistics Panel

- Displays filtered flight counts
- Status classification (on-time/delayed/cancelled counts)
- Average flight distance and extreme routes

### 2.4 Future Development

- Live Data Integration: API-based real-time updates
- Advanced Charts:
  - Scatter plots for delay-distance analysis

## 3. Team Contributions

### 3.1 Yin (Team Lead)

- Designed system architecture and tech stack
- Implemented cross-page state management
- Performed 85% of code reviews/merges
- Optimized list rendering performance (3× smoother scrolling)
- Led team meetings and task coordination

### 3.2 Ye (Data Specialist)

- Designed efficient data structures
- Implemented date range filtering with error handling
- Developed data preprocessing tools
- Fixed 8 critical data parsing bugs

### 3.3 Zhu (UI/UX Designer)

- Created all visual assets (32 custom icons)
- Designed color schemes and style guide
- Developed interface animation system
- Produced 3-minute promotional video

### 3.4 Yuan (Frontend Developer)

- Implemented 5 page layouts
- Developed page transition animations
- Optimized touch interactions
- Completed mobile adaptation

### 3.5 Sun (Visualization Engineer)

- Developed heatmap generation algorithm
- Processed geographic coordinate conversion

- Implemented route density calculations
- Optimized map rendering performance

3.6 Gao (Interaction Developer)

- Assisted heatmap development
- Processed coordinate conversion
- Optimized interaction performance
- Implemented dynamic floating panel

Github history:

main

All usersAll time

Commits on Apr 9, 2025

Add files via upload

Runchen-Ye authored 2 hours ago

Verifieda1bad43

Commits on Apr 7, 2025

Add files via upload

TommyG0606 authored 2 days ago

Verifiedd8b75f5

完整的筛选功能

Mingqi-Yuan authored 2 days ago

Verified28bf048

Commits on Apr 1, 2025

Add files via upload

Han-ZhuAI authored last week

Verifiedf516f66

Commits on Mar 27, 2025

Update Week10\_Lateness\_and\_Airport.pde

Runchen-Ye authored 2 weeks ago

Verified0726f14

Rename Week10\_Lateness\_and\_Date.pde to Week10\_Lateness\_and\_Airport.pde

Runchen-Ye authored 2 weeks ago

Verified1da33ae

syin week 10 main structure (second page change)

Yovi491 authored 2 weeks ago

Verifiede2b3477

Add files via upload

Runchen-Ye authored 2 weeks ago

Verified3c96c7f

Commits on Mar 26, 2025

Add files via upload

XuboSun624 authored 2 weeks ago

Verified584e1ad

Commits on Mar 26, 2025

Add files via upload

XuboSun624 authored 2 weeks ago

Verified584e1ad

Add files via upload

XuboSun624 authored 2 weeks ago

Verifieda3845b2

Commits on Mar 25, 2025

Add files via upload

TommyG0606 authored 2 weeks ago

Verified111832b

Commits on Mar 24, 2025

Create week 10 main structure (第二界面改动)

Yovi491 authored 2 weeks ago

Verifiedfa7d3f6

Commits on Mar 23, 2025

Create Program Structure (完整的page2)

Mingqi-Yuan authored 2 weeks ago

Verifiedfe552a2

Commits on Mar 20, 2025

Add files via upload

Runchen-Ye authored 3 weeks ago

Verifiedd6b8de7

Add files via upload

Runchen-Ye authored 3 weeks ago

Verifiede769dce

Create basici structure(图表版)

Yovi491 authored 3 weeks ago

Verified9572f3b

Update structure main (加航班列表)

Mingqi-Yuan authored 3 weeks ago

Verified2c02ac9

Create structure main (加航班列表)

Mingqi-Yuan authored 3 weeks ago

Verified48ec27b

|                                  |          |         |  |
|----------------------------------|----------|---------|--|
| Commits on Mar 18, 2025          |          |         |  |
| Create structure main(加搜索栏)      | Verified | bf362d9 |  |
| Yovr491 authored 3 weeks ago     |          |         |  |
| Commits on Mar 17, 2025          |          |         |  |
| 热力图2                             | Verified | 694ca44 |  |
| XuboSun624 authored 3 weeks ago  |          |         |  |
| 柱状图2                             | Verified | 212edc8 |  |
| XuboSun624 authored 3 weeks ago  |          |         |  |
| Update Program basic structure   | Verified | 115262b |  |
| Mingqi-Yuan authored 3 weeks ago |          |         |  |
| Add files via upload             | Verified | 0766f70 |  |
| TommyG0606 authored 3 weeks ago  |          |         |  |
| Delete sketch_250313a.pde        | Verified | e1f6eea |  |
| TommyG0606 authored 3 weeks ago  |          |         |  |
| Add files via upload             | Verified | e868a7a |  |
| XuboSun624 authored 3 weeks ago  |          |         |  |
| Add files via upload             | Verified | 3a28eec |  |
| XuboSun624 authored 3 weeks ago  |          |         |  |
| Add files via upload             | Verified | ec71d86 |  |
| XuboSun624 authored 3 weeks ago  |          |         |  |
| Add files via upload             | Verified | a085b1  |  |
| XuboSun624 authored 3 weeks ago  |          |         |  |
| Commits on Mar 16, 2025          |          |         |  |
| Create structure (含数据)           | Verified | ef02fbf |  |
| Yovr491 authored 3 weeks ago     |          |         |  |
| Add files via upload             | Verified | ec71d86 |  |
| XuboSun624 authored 3 weeks ago  |          |         |  |
| Add files via upload             | Verified | a085b1  |  |
| XuboSun624 authored 3 weeks ago  |          |         |  |
| Commits on Mar 16, 2025          |          |         |  |
| Create structure (含数据)           | Verified | ef02fbf |  |
| Yovr491 authored 3 weeks ago     |          |         |  |
| Commits on Mar 13, 2025          |          |         |  |
| Add files via upload             | Verified | fb3c3bf |  |
| TommyG0606 authored last month   |          |         |  |
| Add files via upload             | Verified | 034899e |  |
| TommyG0606 authored last month   |          |         |  |
| Create 3.13                      | Verified | 4ef9fb8 |  |
| Yovr491 authored last month      |          |         |  |
| Create Program basic structure 2 | Verified | 7fb1b2d |  |
| Mingqi-Yuan authored last month  |          |         |  |
| Add files via upload             | Verified | 194499c |  |
| Runchen-Ye authored last month   |          |         |  |
| Create Program basic structure   | Verified | b820d2b |  |
| Mingqi-Yuan authored last month  |          |         |  |
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| Commits on Mar 13, 2025         |          |         |
| Add files via upload            | Verified | 3a65cf4 |
| Han-ZhuAI authored last month   |          |         |
| Add files via upload            | Verified | d885147 |
| Mingqi-Yuan authored last month |          |         |
| Add files via upload            | Verified | 747849f |
| Han-ZhuAI authored last month   |          |         |
| Add files via upload            | Verified | e6ee8f2 |
| Mingqi-Yuan authored last month |          |         |
| Add files via upload            | Verified | f308b67 |
| Han-ZhuAI authored last month   |          |         |
| Add files via upload            | Verified | 498258d |
| Mingqi-Yuan authored last month |          |         |
| Commits on Mar 12, 2025         |          |         |
| Add files via upload            | Verified | c1e67f3 |
| Yovr491 authored last month     |          |         |
| Commits on Mar 11, 2025         |          |         |
| Create week8                    | Verified | 2b822d9 |
| Yovr491 authored last month     |          |         |
| Create 读取数据                     | Verified | 4862c88 |
| Yovr491 authored last month     |          |         |
| Add files via upload            | Verified | d411652 |
| Yovr491 authored last month     |          |         |

Username and Name:

Yovr491—SHIRAN YIN  
 XuBoSun624—XUBO SUN  
 TommyG0606—ZHIHAN GAO  
 Han-ZhuAI—HAN ZHU  
 Runchen-Ye—RUNCHEN YE  
 Mingqi-Yuan—MINGQI YUAN

## 4. System Screenshots

```
// Destination input box (Filter based on the destination city in the file, corresponding to the data in column I)
text("Destination:", 80, 190);
drawInputField(190, 175, 200, 30, destinationText, focusedField == 2);

text("Date Range", 80, 230);

// Fly Date Range Input: three parts (Month, Day, Year), used to filter the file flDate (column A)
text("Fly Date Range:", 80, 255);
drawInputField(225, 240, 40, 30, flyDateLeft, focusedField == 3);
drawInputField(275, 240, 40, 30, flyDateMid, focusedField == 4);
drawInputField(325, 240, 70, 30, flyDateRight, focusedField == 5);

// Arrive Date Range segmented input: three parts, used to filter the arrival date
text("Arrive Date Range:", 80, 280);
drawInputField(250, 270, 40, 30, arriveDateLeft, focusedField == 6);
drawInputField(300, 270, 40, 30, arriveDateMid, focusedField == 7);
drawInputField(350, 270, 70, 30, arriveDateRight, focusedField == 8);

text("Lateness", 80, 320);
drawLatenessButtons();

drawChartButtons();
}

void drawLatenessButtons() {
    drawToggleButton(80, 350, "On-time", onTimeSelected);
    drawToggleButton(200, 350, "Delayed", delayedSelected);
    drawToggleButton(320, 350, "Cancelled", cancelledSelected);
}

void drawChartButtons() {
    drawToggleButton(520, 350, "Heatmap", heatmapSelected);
    drawToggleButton(640, 350, "Route Map", routeMapSelected);
}

void drawFlightInfoPanel() {
    // Call the filter function every time you redraw to ensure that the flight list is updated synchronously when the status and date conditions change
    filteredFlights = filterFlights();

    drawPanel(flightPanelX, flightPanelY, flightPanelW, flightPanelH, "");
    clip(flightPanelX, flightPanelY, flightPanelW, flightPanelH);

    int startIndex = scrollY / flightItemHeight;
    int endIndex = min(filteredFlights.size(), startIndex + flightPanelH / flightItemHeight + 1);

    // Record the flight the mouse is hovering over
    Flight hoveredFlight = null;

    for (int i = startIndex; i < endIndex; i++) {
        Flight f = filteredFlights.get(i);
        int yPos = flightPanelY + (i * flightItemHeight - scrollY);
        String flightInfo = f.flDate + " " + f.mktCarrierFlNum + " " + f.originCity + " to " + f.destCity;

        // If the mouse is in the display area of the current flight, record the flight and change the background color
        if (mouseX > flightPanelX && mouseX < flightPanelX + flightPanelW &&
            mouseY > yPos && mouseY < yPos + flightItemHeight) {
            fill(200);
            hoveredFlight = f;
        } else {
            fill(255);
        }
    }
    rect(flightPanelX, yPos, flightPanelW, flightItemHeight);
    fill(0);
    textAlign(LEFT, CENTER);
    text(flightInfo, flightPanelX + 5, yPos + flightItemHeight / 2);
}
```

```

void drawPage2() {
    textFont(createFont("Arial", 14));
    fill(255);
    textSize(14);
    textAlign(CENTER);
    text("Detailed Flight Information", width / 2, 30);

    int leftMargin = 50, topMargin = 50, panelWidth = 340, panelHeight = 180, gap = 20, subSpacing = 25;
    int subStartOffset = 45;

    drawPanelPage2(leftMargin, topMargin, panelWidth, panelHeight, "Airline Information");
    int subY = topMargin + subStartOffset;
    textAlign(LEFT);
    text("Flight Date: " + selectedFlight.flDate, leftMargin + 10, subY);
    text("Carrier: " + selectedFlight.mktCarrier, leftMargin + 10, subY + subSpacing);
    text("Flight Number: " + selectedFlight.mktCarrierFlNum, leftMargin + 10, subY + 2 * subSpacing);

    drawPanelPage2(leftMargin + panelWidth + gap, topMargin, panelWidth, panelHeight, "Origin Information");
    subY = topMargin + subStartOffset;
    drawWhiteButton(leftMargin + panelWidth + gap + 10, subY - 10, "Airport Code: " + selectedFlight.originCode);
    text("City & State: " + selectedFlight.originCity, leftMargin + panelWidth + gap + 10, subY + subSpacing);
    text("State Abbreviation: " + selectedFlight.originState, leftMargin + panelWidth + gap + 10, subY + 2 * subSpacing);
    text("Region Code: " + selectedFlight.originWac, leftMargin + panelWidth + gap + 10, subY + 3 * subSpacing);

    int secondRowTop = topMargin + panelHeight + gap;
    drawPanelPage2(leftMargin, secondRowTop, panelWidth, panelHeight, "Dest Information");
    subY = secondRowTop + subStartOffset;
    drawWhiteButton(leftMargin + 10, subY - 10, "Airport Code: " + selectedFlight.destCode);
    text("City & State: " + selectedFlight.destCity, leftMargin + 10, subY + subSpacing);
    text("State Abbreviation: " + selectedFlight.destState, leftMargin + 10, subY + 2 * subSpacing);
    println("Debug - destWac = " + selectedFlight.destWac);
    text("Region Code: " + selectedFlight.destWac, leftMargin + 10, subY + 3 * subSpacing);
}

void drawPage5() {
    fill(255);
    textSize(32);
    textAlign(CENTER);
    text("Flight Routes Map", width / 2, 50);

    if (routeMapImage != null) {
        image(routeMapImage, 50, 80, width - 100, height - 160);
    } else {
        text("Error loading Route Map image.", width / 2, height / 2);
    }

    drawBackButton();
}

```

## 5. Conclusion

This system delivers efficient flight data querying and visualization through modular development. Team members contributed specialized expertise to create a fully-featured, user-friendly application. Future work will focus on enhanced data analysis and real-time functionality.